Spring 2021 / EC571

Instructor:Ajay Joshi

Mid-Term Exam 1

Date: February 24, 2021

Name : _____

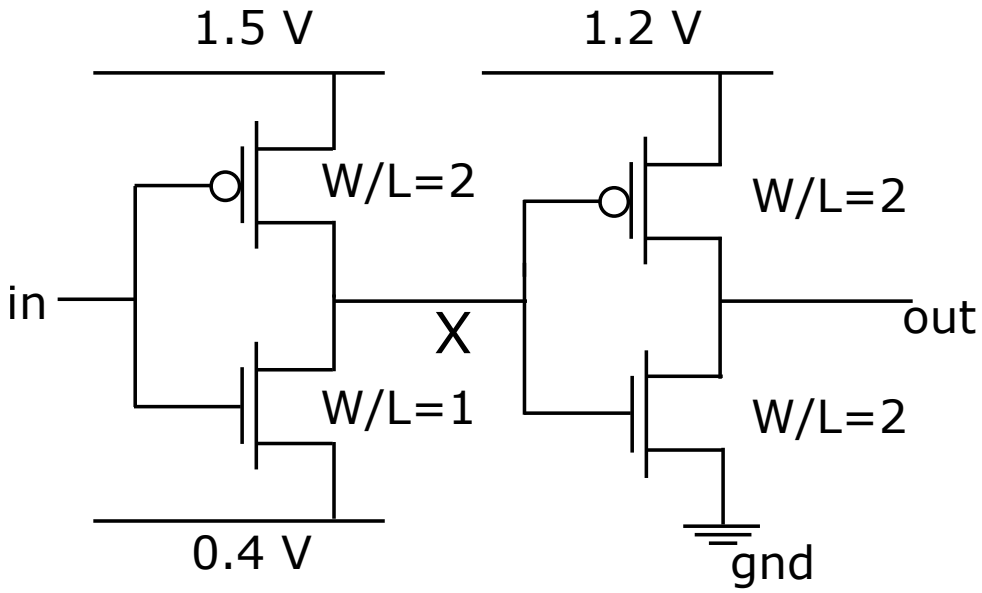
BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
Bonus	5	
Total	100	

1. The duration of the exam is 1 hour 45 minutes.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 6 questions. Please indicate the question number that should not be graded.
5. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
6. Material sharing is not allowed.
7. Calculators are allowed.
8. The exam is open book open notes.
9. Boston University Academic Code of Conduct applies.
10. BU ENG is collaborating with several online services, including Chegg, to monitor activity during exams and obtain account information and IP addresses connected to uploaded exam materials and their access. The penalties for cheating are extremely severe including expulsion from BU.

Question 1

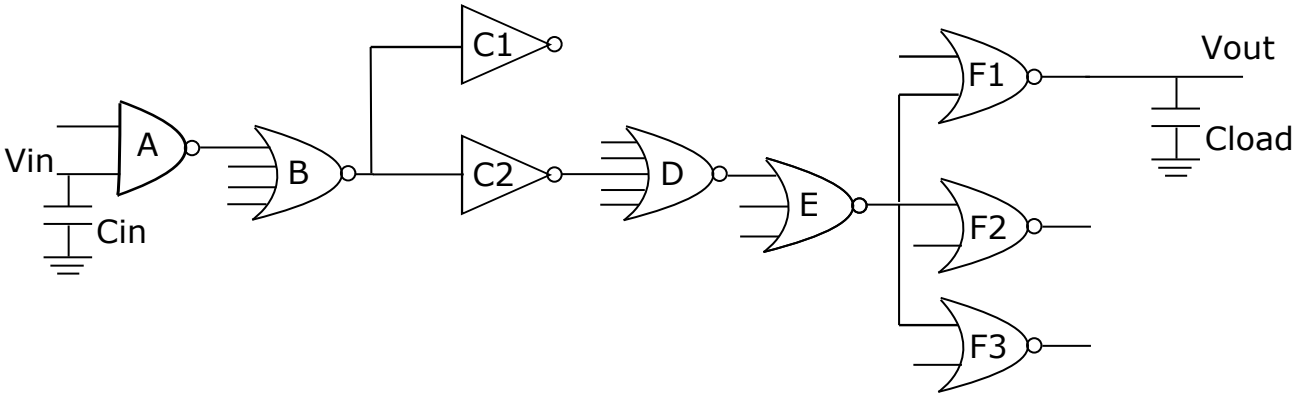
1. Using the simple switch model of the transistors with $R_{sq,NMOS}$ = 13 Kohm/sq, $R_{sq,PMOS}$ = 26 Kohm/sq, $V_{th,N}$ = 0.25 V, $V_{th,P}$ = -0.25 V and V_{DD} = 1.5 V, draw the VTC of the circuit shown below. Assume V_{in} varies from 0 to 1.5V. Determine the values for V_{OH}, V_{OL}, V_{IH} and V_{IL}. Explain your answer in detail. (10 points)



2. What is the region of operation for all four transistors when V_{in} = 1.5V. Assume long channel transistors. (5 points)

3. For the same circuit how much energy is supplied by the power supply V_{DD} to charge a load capacitor of 10 fF at the output when we feed a step input (1.5V $\rightarrow$ 0 V)? (5 points)

Question 2



For the circuit shown in the above figure, using the logical effort technique, calculate the input width of transistors for each gate in the path such that total delay of the path is minimized. You can assume that gates C1 and C2 are identical and gates F1 and F2 are identical. Assume C_{in} = 10 fF, C_G = 1 fF/μm, C_{load} = 120 fF and $2R_{sq(NMOS)} = R_{sq(PMOS)}$. (20 points)

Question 3

Draw the static complementary CMOS implementations of the following Boolean expressions:

1. $F1(A,B,C,D) = \overline{A \cdot B + C \cdot D}$ (5 points)

2. $F2(P,R) = P \cdot R$ (5 points)

3. $F3(K,W,X,Y,Z) = \overline{K \cdot W + X \cdot Y \cdot Z}$ (5 points)

4. $F4(E,F) = E \oplus F$ (5 points)

Question 4

1. Draw the static complementary CMOS logic implementation for the following function - $f(A,B,C,D,E) = \overline{(A \cdot B \cdot C + D)} \cdot E$. (6 points)

2. Determine the widths of the various transistors as a function of W so that worst case $\tau_{HL} = \tau_{LH}$ and the delay matches with the delay of an inverter. You can assume $2.5R_{sq(NMOS)} = R_{sq(PMOS)}$. Ignore the internal capacitances of the non-output nodes while determining the transistor sizing. (7 points)

3. Calculate the worst-case τ_{HL} and τ_{LH} for both gates using the following parameters – width of the NMOS in the inverter $W = 0.5$ micron, $L = 0.25$ micron $R_{sq(NMOS)} = 10$ Kohm/sq, $R_{sq(PMOS)} = 25$ Kohm/sq, $C_G = 1.0$ fF/μm, $C_D = 1.2$ fF/μm, $C_L = 75$ fF. Include the internal capacitances of the non-output nodes while determining the delay. (7 points)

Question 5

You have to design the static complementary CMOS implementation of a 1-bit adder for minimum delay (assuming a fanout i.e. stage effort of 4) and driving a load of 10 fF. As part of the design you need to determine the width of all transistors. You can use the following transistor parameters for the design - L = 0.25 micron $R_{sq}(NMOS)$ = 13 Kohm/sq, $R_{sq}(PMOS)$ = 26 Kohm/sq, C_G = 1 fF/micron, C_D = 1.5 fF/micron. You can assume that you have both complemented and non-complemented inputs (i.e. you have $A, \overline{A}, B, \overline{B}, Cin, \overline{Cin}$ and do not need additional inverters to generate these signals) and that input signals have very small rise and fall times.

1. Determine the Boolean expression for the sum output and carry output in it's Sum-of-Products form. (5 points)

2. Draw the gate-level schematic using the Boolean expressions for the sum output and carry output, and determine the longest paths. (5 points)

3. Using the Logical Effort method, determine the size of all the transistors in the design for the sum output logic and the carry output logic. (10 points)

Question 6

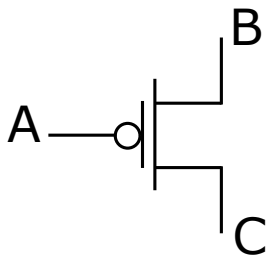
1. What are the different regions of operation of an NMOS transistor? Write the current equations for each one of the different regions of operation. Assume long-channel transistor. (6 points)

2. Write the current equations for each one of the different regions of operation if the NMOS transistor was a short-channel transistor. (3 points)

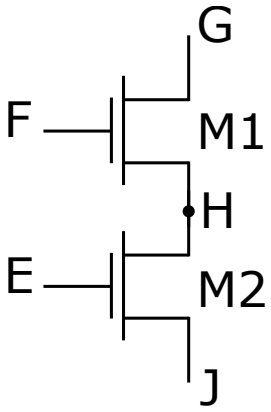
3. What is the Miller Capacitance effect as it applies to CMOS inverters? (3 points)

4. Explain the velocity saturation effect in your own words. (2 points)

5. For the PMOS transistor shown below, if terminal A is connected to *GND*, terminal B is connected to *GND* and terminal C is connected to V_{DD} = 2.5V, then what is the region of operation of the transistor if $V_{DSAT,p}$ = -1.0V and $V_{TH,p}$ = -0.4V? (2 points)



6. For the two NMOS transistors shown below, if terminal E is connected to *GND*, terminal F is connected to V_{DD} = 2.5V, terminal G is connected to V_{DD} = 2.5V, and terminal J is connected to *GND*, what is the voltage at node H, and what are the regions of operation for both NMOS transistors if $V_{DSAT,n}$ = 0.6V and $V_{TH,n}$ = 0.4V? (4 points)



Bonus question (5 points)
Prof. Joshi was planning to go for Tokyo Olympics last year, but could not go as the Tokyo Olympics were postponed to 2021. Please list 5 Olympic sports that he should definitely attend if he is able to go to Tokyo Olympics.